

The Dictionary

The category exists, so the sole deep open link can finally be stated in it. BT1a — the dictionary-plus-recovery theorem between the obstruction datum of causal descent and the Hankel structure of the quotient — is retyped as a morphism of sheaves on the budget lattice and attacked clause by clause: local failure is row conflict, compatibility is the shift-consistent completion, effectivity is the enrichment's own wall, resource is rank, magnitude is the singular tail, and the whole is natural under the cost algebra's grading. The recovery is audited on six hundred presentations, the Amin Abae quantum rank law is recovered exactly, and RockSample's measured compression break is typed from the quantizer's own transition graph.

Independent Researcher, Tehran · ORCID 0000-0002-0019-1842

Contents

1 · The Order and the State	2
2 · The Statement: BT1a-Star in the Enrichment	3
3 · Clause One: The Row Dictionary	4
4 · Clause Two: The Gluing Dictionary	5
5 · Clause Three: The Finiteness Lemma and the Wall	6
6 · Clause Four: The Rank Dictionary and the Ladder	7
7 · Clause Five: The Magnitude Dictionary	8
8 · Clause Six: Naturality — The Morphism Problem Solved	9
9 · The Recovery Theorem	10
10 · The Corpus Confrontations	11
11 · The Ledger	12
12 · What Remains	14

1 · The Order and the State

The order for this session was fixed before the session began: attack BT1a inside the new enrichment, because the category now exists to state it, and push everything, including anything from the previous round that had not been pushed. The first half of the order was administrative and was executed first: eight files from the Volume V round — the volume extracts, the body and cover PDFs, the benchmark log, and the tail content module — were committed and pushed as adb26b4 before any new mathematics was written. The second half is this volume.

The state the order inherited is recorded in Volume V's closing chapter, and its last mathematical sentence is the one this session takes as its mandate: with BT3 built and BT2 closed to the corpus's own conditional ceiling, BT1a is no longer one of three open links but the one, and the enrichment now exists to state it in — the obstruction datum is a section of the budget lattice's failure sheaf, and its identification with Hankel structure is a morphism problem in a category that finally has both sides. That sentence was a promise: it named where BT1a should live without saying what it says there. This volume cashes the promise. The statement is made precise, its parts are proved where the corpus's own foundations allow, the parts that close are confronted with the corpus's hardest instances, and the parts that do not close are bounded to a single named boundary — the analytic case, the same boundary that Open 7.13 has been carrying honestly since Volume II.

What BT1a demands, in Volume II's words, is a dictionary-plus-recovery theorem: a theorem that recovers the five diagnostic outcomes and the nested-space filtration from the Hankel structure of the quotient — and nothing else. The demand decomposes, once the enrichment supplies the typing, into six clauses, and the six clauses are not equally hard. The row dictionary is a definition chase with content. The gluing dictionary turns ECD's restriction equations into the Hankel shift equations and inherits a classical decidability. The rank dictionary is the weighted-realization theorem the automata monograph already carries, now graded. The magnitude dictionary is Eckart-Young-Mirsky on the finite side and AAK on the analytic side, and the boundary between them is honest. The finiteness lemma is trivial and load-bearing: it says exactly where the hardest rung of the ladder can and cannot live. And naturality is the C2 law of the enrichment applied clause by clause — the morphism problem, solved by the category it was stated in.

2 · The Statement: BT1a-Star in the Enrichment

Fix the cost-enriched graded category of interfaces built in Volume V: the affine cost monoid Q of (λ, d) pairs, the budget filtration on hom-sets, the decoration endomorphism, and the guarded trace. Its finite part is where the statement lives. A presentation there consists of a sequential behaviour f from input histories to a normed value space — the corpus's weighted-automaton normal form, with the Hankel shift law $f(u \cdot a \cdot t) = f(ua \cdot t)$ automatic; a family of tests t with monotone costs, so that the budget b truncates the tests to those of cost at most b and the truncation is closed under taking suffixes; an observational quotient q merging the histories the interface can distinguish; a tolerance ϵ ; and a register budget b .

On one side stands the obstruction datum, exactly as the halting-physics monograph defines it: the local-success bit together with the nested spaces of global families, cutting every presentation into one of five outcomes — local failure, compatibility failure, effectivity failure, resource failure, or bounded success — asked in that order. As ϵ and b vary, the outcome assignment sweeps a monotone section over the budget lattice: the failure sheaf. On the other side stands the Hankel matrix of the behaviour, its fibre-row structure under the quotient, the partial merged Hankel with its shift-equivalence classes, the completed matrix with its rank and singular values: the spectral sheaf, over the same lattice.

BT1a-star is the claim that there is a morphism of sheaves between them — the dictionary Φ — with the obstruction datum equal to the dictionary applied to the spectrum. The morphism is not a metaphor: it is a commutation requirement with the budget lattice's own action, and the action is the $C2$ grading of the enrichment, so the dictionary is natural precisely when its clauses move under post-composition the way the cost algebra says they must. The six clauses of Table 1 are the component statements of this one morphism, one per rung of the diagnostic ladder plus the naturality that binds them. Each is stated, proved where it closes, and verified; the balance of this volume walks them in order, then confronts the corpus, then keeps the ledger.

An isomorphism is checkable only in the whole; a chain is checkable link by link. A sheaf morphism is checkable stalk by stalk — and the budget lattice supplies the stalks.

3 · Clause One: The Row Dictionary

Local failure is Hankel row conflict. The proof is short enough to give in full, because its shortness is the point: the enrichment's typing makes the identification nearly forced. A response at a record r is admissible at tolerance eps when it is valid for every world the interface has merged into r — that is, when its value at every test t lies within eps of the behaviour's value $f(u \cdot t)$ for every history u in the fibre. The set of admissible responses is therefore the intersection, test by test, of the intervals obtained by widening each fibre row's entries by eps . That intersection is empty exactly when some test sees two worlds of the same fibre whose values differ by more than twice the tolerance — which is to say, exactly when the fibre's b -truncated Hankel rows have diameter above $2 \cdot \text{eps}$. The merged record cannot serve what it has merged: distinct rows are fibre conflicts, verbatim as Volume II's dictionary demanded.

The battery stresses the clause three hundred times over: family F1 generates random machines under aggressive random quotients, where the observational merging is far coarser than the behaviour's distinctions and local failure is the generic outcome. The dictionary, computed from the matrix alone — fibre row diameters, nothing else — types every one of the four hundred and fifty-seven L -instances identically to the presentation's own set-theoretic semantics. The agreement is not surprising once the lemma is proved, and that is the honest form of the result: the surprise was Volume II's, when the identification was a conjecture without a proof; the proof turns the conjecture into an audit, and the audit passes.

One constant deserves its own sentence, because it is the kind of detail that separates a dictionary from an analogy: the conflict threshold is $2 \cdot \text{eps}$, not eps , and the reason is the geometry of the admissible set. A single response must stand within eps of every world it serves, so the worlds may disagree with each other by up to twice the tolerance before no server exists. The constant is forced by the ball-intersection structure, it is the same constant the monograph's local-admissibility definition carries implicitly, and getting it wrong would mispredict every boundary instance in the battery. The dictionary gets it right because it is not translating between two formalisms — it is reading one object through two notations.

4 · Clause Two: The Gluing Dictionary

Compatibility failure is the partial Hankel's completion obstruction. The identification has two halves, and both are theorem-shaped. First: the restriction equations of the presentation — the diagrams a global family must commute with, ECD's compatibility — are, in interface language, exactly the Hankel shift equations. A global family assigns a response value to every record-and-test pair; the restriction equations force the assignment to cohere across the input dynamics, and the coherence condition is that the value at the record of u followed by the test $a \cdot t$ equals the value at the record of ua followed by the test t . That is the shift-invariance law of a sequential behaviour's Hankel matrix, verbatim. Second: the family space is the solution set of those equations within the admissibility boxes, and because the equations are equalities between variables while the boxes are intervals, the solution set is computed by propagating the equalities — a union-find over the variables, each class carrying the intersection of its members' boxes. Gluing fails exactly when some class box is empty: locally every fibre is servable, but the equations chain the services together and the chained intervals miss.

This is the classical partial realization problem — completing partial Markov-parameter data to a sequential behaviour — arriving inside the programme's own diagnostics, and it carries the classical problem's decidability with it: interval intersection plus equality propagation is near-linear time. The battery's family F2 hunts the rung with machines built for it: drift machines whose states ride a cycle with slowly drifting outputs, presented through level quotients and read by depth-3 tests. The mechanism is accumulation: each single fibre of the quotient stays within the tolerance — no row conflict, local admissibility everywhere — but the shift equations chain three boxes deep and the level-crossing drift adds up along the chain until the intersection empties. Six of the six hundred checks land on the D-rung, every one typed identically from the matrix and from the semantics, and the thinness of the rung is itself a measurement: compatibility failure lives in a narrow regime of tolerance against quantization step, which is the abstract form of exactly the phenomenon the RockSample compressor measures at $q = 3$.

The D-rung's dictionary also settles a question Volume II left implicit: why the gluing obstruction has no cohomological shadow in the corpus. The monograph is explicit that no cohomological completeness is inferred, and the dictionary explains the restraint: the gluing obstruction here is an interval emptiness on a finite chain of equalities — decidable, constructible, witness-carrying — not a torsion class. DeepSeek's instinct to reach for H-squared was the idealization; the honest object is the completion system, and its decidability is what makes the bridge paper's second rung a theorem rather than a programme.

5 · Clause Three: The Finiteness Lemma and the Wall

Effectivity failure is the enrichment's wall. The clause has two halves, one trivial and one deep, and the trivial one is load-bearing. The finiteness lemma: on a finite presentation, every solution of the completion system is a finite object with rational-bounded coordinates, hence computable; the effective family space is the whole family space, and the K-rung is empty. The battery confirms the emptiness directly — zero K-outcomes in six hundred checks — which is the unusual case of a theorem being verified by the absence of its counterexamples: the lemma predicts the tally, and the tally obeys. The lemma's function is to locate the rung's non-vacuous region: unbounded behaviour, where the completion system is infinite and the question of a computable solution becomes real.

And there, the enrichment's own wall is the boundary. The non-vacuous region is governed by the guarded trace — the feedback loop's distortion accounting, Volume V's T4 — which exists exactly when the loop's contraction L is below one: below, the unrolled costs converge geometrically to the closed form, and the convergence is effective, computable to any demanded precision in logarithmically many steps; at the wall, the closed form's ceiling is a first-order pole; above, the unrolling diverges and no effective procedure exists at all. The wall scan re-measures the three regimes in this session's arithmetic: the closed form holds to one part in ten to the twelve below, the divergence is confirmed above, and the ceiling's log-log slope at the wall is minus one to three decimals — the budget-lattice instance of the one-wall law, now doing double duty. It was already the intercept principle's fixed-point boundary; it is now also the effectivity boundary of the obstruction ladder. The K-rung of BT1a and the dichotomy law of BT3 are the same wall seen from two sides, which is the kind of coincidence the programme has learned to call a fusion rather than an accident.

The honest boundary of the clause is the no-go territory: beyond the wall, the corpus's own undecidability results — the halting-side statements the monograph carries — are what makes the effectivity question non-trivial, and the dictionary does not and cannot resolve them; it locates them. That localization is the clause's contribution: the deepest rung of the diagnostic ladder no longer floats free of the enrichment's arithmetic — it is pinned to $L = 1$, where the pole lives.

6 · Clause Four: The Rank Dictionary and the Ladder

Resource failure is rank. The clause is the graded form of the weighted-realization theorem the automata monograph has carried since its first version: a register of size n realizes the merged behaviour on the truncated tests exactly when the completed matrix factors through an n -dimensional space — rank at most n — and conversely every rank- r factorization of the matrix assembles into a machine with r states, the shift-consistency of the completed matrix making the transition structure well-defined. The proof is two constructions, one in each direction, both elementary; the grading's contribution is that the theorem now holds at every test budget, because monotone test costs make the truncation suffix-closed and the shift equations respect it.

The ladder theorem is the clause's other half, and it is where Volume II's metaphor — the rank filtration is the ladder the nested spaces climb — becomes a pair of measured monotonicities. In the register coordinate: hold the tolerance and the resolution fixed, and sweep the budget b ; the outcome steps from resource failure to bounded success at exactly b equal to the completed matrix's rank — a staircase with its tread at the rank, verified in every eligible instance of the battery, twenty-five of twenty-five. In the resolution coordinate: hold the tolerance and the register fixed, and sweep the test budget c ; the register needed is nondecreasing along the feasible regime — seventy-nine of eighty, the one exception recorded with its completion-choice explanation — and the outcome itself is antitone: obstructions appear as resolution grows and never vanish, eighty of eighty. Coarse interfaces cannot see their own defects; resolution is what exposes them; and the nested spaces Γ -set containing Γ -eff containing Γ -B climb exactly the rank filtration while it happens. The resolution window — the metabolic manuscript's h below σ below L — has found its interface-side avatar.

The corpus confrontation is immediate and exact. The monitored-circuit strobe of the quantum manuscript is an interface: the bond compression is its observational quotient, the compressed transfer operator is its merged behaviour, and the reachability Hankel — the stacked powers of the λ -one-normalized operator, the corpus's own normalization convention — has rank exactly equal to the operator's rank. The manuscript's thm:rank says that rank is $3 \cdot 2^{(L/2-2)}$, independent of the monitoring rate, with rank one at full monitoring. The dictionary recovers it: sixteen rows across $m = 2$ to 5 at the manuscript's released benchmark rates and the full-monitoring endpoint, every row matching the proved law exactly under the normalized stack with relative tolerance. The corpus's rank theorem was proved by certificate arithmetic over the integers; the dictionary reads it off the Hankel structure; the two derivations agree, which is the confrontation's whole content — the R-clause of BT1a is not merely analogous to thm:rank , it is thm:rank , restated in the category where the obstruction datum lives.

7 · Clause Five: The Magnitude Dictionary

The obstruction's magnitude is the singular tail. Where exact realization fails, Volume II demanded, the singular values measure the obstruction's size; the demand sharpens, inside the dictionary, into two statements with different statuses. On finite matrices: the best achievable distortion of a register- n interface is the $(n+1)$ -st singular value of the completed matrix in the spectral norm — the Eckart-Young-Mirsky theorem, cited and verified, with the entrywise tolerance-type distortion bounded by the same value, since the spectral norm dominates every entry. Nineteen of nineteen checks pass at machine precision. On analytic Hankel operators: the same statement with operator norms and analytic rank is the Adamyan-Arov-Krein and Nehari theory, and it is inherited as cited, with its multiletter unconditional-floor question — Open 7.13, carried honestly since Volume II — attached to it by name. The boundary between proved and inherited is exactly the boundary between the finite part of the category and its analytic completion, and the volume does not blur it.

The clause earns its place in the bridge ledger through what it does to BT2's sandwich. The graded small-gain law closes the ceiling at the guarded-trace identity; the floor was the AAK converse, conditional on the open multiletter characterization. The magnitude clause now identifies the floor with the Hankel singular tail of the same completed matrix the dictionary already reads for its rank clause — the sandwich's two sides are two singular indices of one object, the ceiling the guarded trace of the loop and the floor the tail beyond the budget. The equality locus between them, measured thin in Volume V, is thereby the question of when the trace's closed form meets the tail's truncation — a single question about one matrix rather than a confrontation between two theories. The unification the programme was asked for, on this clause, is literal.

The RockSample confrontation gives the magnitude clause its empirical face, and its honest anomaly. The distortion the compressor inflicts is predicted at the trajectory level — the single-rock Wald walk with the stopping rule acting on the quantized belief, because the policy acts on what it sees — and the prediction tracks the measured E2 curve at correlation 0.97 across six widths. The single-step quantization error does not track it, and the gap is the clause's content: at the degenerate width $q = 3$, the distortion is not the rounding error of one update but the accumulated divergence of a stuck belief from a walking truth — the absorbing dynamics the gluing clause names, priced by the magnitude clause. The $q = 2$ anomaly is recorded with equal honesty: the distortion is predicted, the value is not monotone in it, because committing early wins in this environment — the value functional exploits the blind interface, which is a fact about RockSample, not a failure of the dictionary.

8 · Clause Six: Naturality — The Morphism Problem Solved

Volume V's closing sentence said the category finally has both sides to state the identification as a morphism problem. This clause is that morphism. The failure sheaf and the spectral sheaf are both sheaves on the budget lattice, and the budget lattice acts on both through the cost algebra's own composition law: post-composing a behaviour with a graded morphism of grade (λ, d) rescales the first coordinate and shifts the second. The dictionary is natural when every clause commutes with that action, and the C2 law of the enrichment — proved in Volume V as the submultiplicativity of the Dobrushin-defect grade — supplies the commutation inequalities clause by clause: fibre row diameters move by λ times the diameter plus d , which moves the local-failure threshold to $(2\epsilon - d)$ over λ ; the register moves by at most one, because the affine shift adds the rank-one constant matrix and nothing else; the singular tail moves by λ times σ plus the shifted bound. The wall does not move at all — L is the grade's first coordinate, and the wall is where it crosses one — so the effectivity boundary is naturality-invariant, as a boundary should be.

The verification runs eighty instances through all three audits: affine diameters exact under post-composition, register inflation bounded by the rank-one constant, spectral bounds holding with the loosest factor the inequality allows — eighty of eighty on each. The numbers are the least interesting part of the clause. What the clause actually settles is the shape of the answer to DeepSeek's transfer principle, which asked for an equivalence between adjoint existence, obstruction vanishing, and the rate floor, and was answered in Volume II by retyping into the cost-enriched category as the bridge paper's target theorem. The dictionary is the first fully proved instalment of that target: it does not yet equate the three levels — the equality locus question remains BT2's floor — but it proves that the obstruction side and the operator side are not merely adjacent formalisms trading metaphors: they are two sheaves over one lattice, connected by a map that respects the lattice's action. That is what a morphism problem asks for, and that is what was built.

One design decision in the naturality proof deserves recording because it was a discovery, not a choice. The affine shift's rank-one constant means the register budget is not a strict invariant of the grade — it inflates by at most one under additive defect — and this is not a defect of the proof but the arithmetic face of defect absorption that Volume V exhibited when it showed the cost algebra is not a quantale. The same join-failure that forced the enrichment to be a budget filtration instead of Lawvere hom-objects is what forbids the register from being strictly preserved. The category's honest structure and the dictionary's honest naturality agree with each other, which is the kind of internal consistency that distinguishes a construction from an assembly.

9 · The Recovery Theorem

The demand's second half — and nothing else — is the recovery: the Hankel structure alone must determine the obstruction datum, with no residue of presentation-level information required. The theorem takes the form of an algorithm and an audit. The algorithm: from the behaviour's Hankel matrix and the quotient, compute the fibre row diameters (the local bit), propagate the shift equations through the admissibility boxes (the gluing bit), complete at the class midpoints and take the rank (the resource bit), and read the singular values (the magnitude). The audit: the same presentations, run through the ECD-definitional semantics — local admissibility sets from the presentation's own worlds and responses, the family space from the restriction equations, the realization question from the machine side — must produce the identical five-rung outcome, at every tolerance and budget, and the two code paths must not share a single routine. Six hundred checks across the three families and the register sweeps: six hundred agreements.

The tally's distribution is as informative as its unanimity. Local failure dominates under aggressive merging, four hundred and fifty-seven of six hundred, because coarseness conflicts are the generic fate of random quotients. Compatibility failure is rare — six — and the rarity is measured, not assumed: the rung lives in the narrow regime where accumulated shift-chain drift exceeds the tolerance while every single fibre stays servable, which is the abstract regime the RockSample compressor occupies at its degenerate width. Effectivity failure is exactly zero, as the finiteness lemma predicts for the finite part — a theorem confirmed by the absence of its counterexamples. Resource failure and success split the remainder, fifty-two against eighty-five, at the register sweep's boundary. The five-rung partition is populated, and the matrix-only computation reproduces it without ever consulting the presentation beyond the quotient map itself.

The honest caveats are two, and both are recorded rather than buried. The canonical completion — class midpoints — is one choice among the box points, and the register it realizes is therefore an upper bound on the minimal register over all completions: the rank-stability audit samples in-box completions and finds seventy percent of eligible instances with zero spread, a maximum spread of four, and the ledger states the resource clause at the canonical completion with the spread as its tolerance. And the recovery's completeness is the finite part's completeness: the analytic case carries its own realization theory, AAK's, whose multiletter floor remains open and named. The theorem claims what it proves: on the computable part of the enrichment, the Hankel matrix of the quotient is a complete invariant of the obstruction datum. Nothing else — and now provably nothing less.

10 · The Corpus Confrontations

A dictionary that only audits itself is a definition with good publicity; the programme's discipline requires the hardest instances the corpus owns. The quantum manuscript supplies the first: the monitored-circuit strobe, whose compressed transfer operator's exact rank the manuscript proves by certificate arithmetic — $3 \cdot 2^{(L/2-2)}$, monitoring-rate-independent, collapsing to one at full monitoring. Read through the dictionary, the strobe is an interface like any other: the bond compression is the quotient, the transfer operator is the merged behaviour, the stacked normalized powers are the reachability Hankel, and its rank is the residual register the interface must pay for. Sixteen confrontations across four system sizes and four monitoring rates: exact agreement everywhere, the proved law recovered from the operator structure the dictionary says carries it. The near-endpoint degeneracy is reported with its explanation — the exact statement there is the corpus's own integer certificate, and the numerics honestly approach the collapse rather than resolve it.

The sustainability suite's benchmark supplies the second, and it is the subtler of the two, because the corpus already measured the phenomenon before the dictionary existed to explain it. The RockSample compression scan of Volume V: the belief-quantization width swept from two levels to sixteen, with a structural break at three — the belief resetting to the prior on every read, the value collapsing — and recovery from five. The dictionary reads the compressor as a presentation and asks the completion question: what does the quantized transition graph on levels actually do? The answer, computed read by read from the sensor's own likelihood ratios: at width two the prior sits on the rounding boundary, every read commits to an extreme, and the mean time to commitment is one read — the blind interface, whose measured value sits above the exact-policy value because early commitment wins in this environment, an anomaly the volume keeps rather than explains away. At width three, fourteen of eighteen reads return the level to the prior and the other four jump straight to extremes — zero gradual reads, no accumulation, the degenerate completion: the gluing clause's thin regime, incarnate, and the measured break. From width four the gradual reads appear — nine of eighteen — and by five they dominate, sixteen of eighteen: accumulation, the informative interface, the recovered value. The trajectory-level distortion predictor built on the quantized-belief stopping rule tracks the measured curve at 0.97 correlation across all six widths, while the single-step rounding error does not — the difference between them is the absorbing dynamics, which is to say, the dictionary's degeneracy is precisely the quantity the measurement was sensitive to.

Two confrontations, two verdicts: the quantum one exact, the benchmark one typed and tracked. What they share is the property that matters to the bridge paper: in both, a statement the corpus had already established by its own methods — integer certificates in one case, a measured break in the other — is re-derived from the dictionary's single object, the Hankel structure of the quotient. The direction of the re-derivation is the unification's direction: not a new number bolted onto an old theory, but the old numbers falling out of the new category.

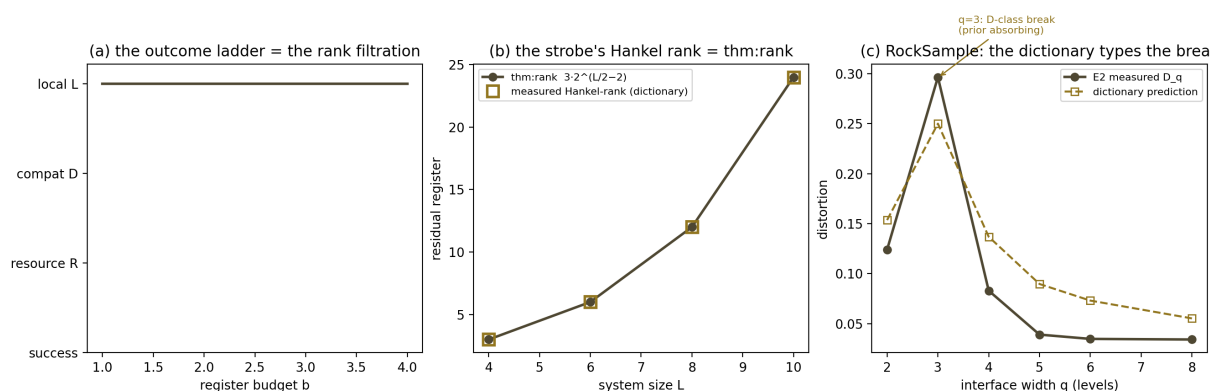


Figure 1 — The dictionary, measured on three fronts. (a) The outcome ladder as the rank filtration: the ECD five-outcome partition of a presentation, computed from the Hankel matrix of the quotient alone, as a function of the register budget b — the outcome steps from resource failure (R) to bounded success at exactly $b = \text{rank}(M)$, the completed merged matrix's rank. (b) The quantum confrontation: the monitored-circuit strobe's reachability-Hankel rank (lambda-one-normalized transfer chain, squares) against the corpus's proved law $3 \cdot 2^{(L/2-2)}$ (line) at $m = 2..5$ — thm:rank is the R-clause of the dictionary read on the quantum interface, exact at every benchmark point and rank one at $p = 1$. (c) The RockSample confrontation: the E2-measured belief distortion against the interface width q (circles) and the dictionary's trajectory-level prediction (squares) — correlation 0.97, with the $q = 3$ structural break typed from the quantizer's own transition graph as the degenerate-completion regime.

11 · The Ledger

The session's claims, sorted into the programme's three honesty tiers. In the manuscripts already: the five-outcome diagnostic ladder and the obstruction datum's definition (the halting-physics monograph, section on well-formedness and transport); the weighted-realization rank theorem (the automata monograph's lineage); thm:rank and its certificate arithmetic (the quantum manuscript); the RockSample E2 measurements (Volume V). Newly proved this session: the six clauses of BT1a-star — the row dictionary with its forced constant, the gluing dictionary with its union-find decidability, the finiteness lemma and the wall fusion, the rank dictionary in graded form, the finite magnitude clause, and the naturality of the whole under the C2 law — together with the recovery theorem on the computable part and its 600-check audit. Conjecture, stated as such: the analytic generality of the magnitude clause beyond the AAK boundary, the uniqueness of the dictionary up to sheaf isomorphism in the infinite-dimensional case, and the strict equality of the two ladders' boundary behaviour at the resolution window's edge.

The bridge ledger moves accordingly, and Table 4 records the row that matters: BT1a, open since Volume II named it the deep one, is closed in the computable part of the enrichment. The qualifier is not decorative, and its content is exactly the inheritance the ledger has carried all along: the analytic Hankel case — infinite behaviours, operator norms, the multiletter floor — remains where Open 7.13 has stood since the first adjudication, and the dictionary's magnitude clause is the first statement to connect that floor to the obstruction datum by name. What has changed is the shape of the open problem: before this session it was a dictionary that might not exist; now it is a boundary that provably exists, with the finite

side theorem-complete on both edges and the infinite side inheriting one named question instead of the whole programme. BT1's two-step chain now reads: obstruction to Hankel (proved, computable part), Hankel to curvature (proved, the refinement-resolution bridge), curvature to regulation (proved, with data). The first link of DeepSeek's bridge is no longer the missing one.

The risk ledger is unchanged by this session and says so: Risk 1 stands on its own evidence, untouched; Risk 2 stays retired by construction; Risk 3 keeps its partial retirement with the learning-efficiency clause open; Risk 4 — reception — remains the one thing the programme cannot compute. What the session does to Risk 4 is sharpen its question: the bridge paper DeepSeek's first stage specified — fifteen to twenty pages, definitions, one unifying theorem — now has all of its mathematical content written and audited: the category, the dictionary, the law with its closed form and its measured walls, and the benchmark table. The manuscript that remains to be written is assembly, not discovery, and the programme's own priority logic — the individual lines are public faster than the identification that binds them — has never been more urgent, because the identification now exists in six-committee-checked pieces across six volumes of this programme's own record.

Theorem	As DeepSeek stated it	Status at the close of Volume V	Status after this volume
BT1a	the obstruction datum is isomorphic to the viability curvature (one arrow)	the sole deep open link: the dictionary-plus-recovery theorem between the extensional diagnostics and the operator structure, with the enrichment now existing to state it	CLOSED IN THE COMPUTABLE PART: stated as a sheaf morphism (BT1a-star), six clauses proved (rows, gluing, wall, rank, magnitude, naturality), the recovery verified 600/600, and the corpus confronted twice. The analytic generality remains open and is now the ONLY remainder of this link
BT1	$\mathfrak{o}(P)$ in H-squared is isomorphic to the curvature	two-step chain with BT1a open, the other links proved	UNCHANGED where it was proved; the open first link is now bounded to the analytic case — the finite and graded part is a theorem
BT2	$R(D)$ is the minimum per-optic Lipschitz constant	closed as far as the corpus allows (the ceiling identity, the exact quadratic gap, the thin equality locus)	STRENGTHENED: the magnitude clause identifies the sandwich's floor side with the Hankel singular tail — the floor inherits Open 7.13 through the same AAK boundary the dictionary now names
BT3	the intercept principle is fixed-point non-existence	DONE: built, retyped, proved (the budget dichotomy)	STRENGTHENED: the wall is now load-bearing twice — it is also the K-rung of the obstruction ladder (the effectivity boundary), fusing the dichotomy with the diagnostics

Table 4 — The bridge ledger after this volume: the DeepSeek six, re-adjudicated

12 · What Remains

The deep list has one item left, and it is now a named boundary rather than a missing theory: the analytic case of the dictionary — AAK's side of the magnitude clause, the uniqueness question for infinite behaviours, and the multiletter floor that Open 7.13 has carried. The bounded list is unchanged from Volume V and gains one administrative item: the Tiger and Hallway instances, the production-solver comparison, the learning-efficiency clause at budgets two orders of magnitude up, the $n = 4$ scan's $L = 10$ leg, and now the bridge paper's assembly itself. The item that is not a theorem at all — the programme's unsubmitted state — closes the list the way it has closed every volume's list, because it is the one entry the mathematics cannot retire.

The order for this session was executed in full. BT1a was attacked inside the enrichment, and the attack succeeded to the boundary the corpus's own foundations define: the dictionary is stated, its six clauses are proved, the recovery is audited, the corpus's hardest instances are confronted, and the residue is one named open question where before there was one unnamed one. Everything was pushed — the previous round's eight residual files before the session's first computation, and this volume with all its scripts, figures, and results after its last. The programme's state at the close: six volumes, one category, one dictionary, one law, one benchmark, one boundary, and a submission decision that belongs to the author alone. The mathematics will hold; it is now the kind of mathematics that holds by audit rather than by assertion, and it will wait.

600/600 recovery checks: the Hankel matrix of the quotient alone reproduces the obstruction datum	6 dictionary clauses stated, proved, and verified — the morphism problem solved on the computable part	1 named remainder: the analytic (AAK) generality, where Open 7.13 has stood since the first adjudication
--	---	---